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INTERNATIONAL QUALIFICATIONS

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International AS and A-level

Physics (9630)
PH04 Energy and Energy resources
Report on the Examination

January 2025

REPORT ON THE EXAMINATION: INTERNATIONAL A-LEVEL PHYSICS (9630) PH04 – JANUARY 2025

General

Numerical questions were better answered than those which required written responses. Whilst, in general, the normal range of extraction is $\pm$ half a square when extracting data from graphs, students should be made aware that this is not always appropriate. This was the case for the graphs in Q4 and Q5 where the data extracted could not be above or below certain values. Students should be reminded that the command word "deduce" in a question means that they are required to draw a conclusion. As such, in parts of Q1 and Q2, students needed to be clear as to whether they agreed or disagreed with the statement.

Question 01

Whilst students were able work out the de Broglie wavelength and access two marks, many failed to make reference to its suitability in measuring the diameter. "3 fm is less than 11 fm" was a common incorrect response. There was often confusion between parts 01.3 and 01.5 with some students suggesting that diameters for the two isotopes would be the same as the charge was the same. Others referred to the different number of nucleons without making a specific reference as to how the numbers were related to the diameter.

Question 02

2.1 Many students were aware that a gas exerts pressure on the walls due to collisions. To score all 4 marks, references to a change in momentum and Newton's laws were essential. There were very few attempts at a full explanation. Most students were able to score 1 for relating pressure, force and area. There were some who used $\boldsymbol{p} = {}^{\boldsymbol{F}}\!/_{\boldsymbol{S}}$ rather than ${}^{\boldsymbol{F}}\!/_{\boldsymbol{A}}$.

2.2 There were some careless errors such as getting the ratios reversed or not taking the square root, but more than 85% of students scored all three marks.

Question 03

3.1 There were some good responses with students showing a reasonably clear understanding of the fission and fusion processes. However, a fair proportion used incorrect terminology such as atoms and molecules instead of nuclei, and collision with the neutron instead of absorption of the neutron to initiate fission.

3.2 Students needed to state that superconducting electromagnets are able to produce strong magnetic fields. A large number of students mentioned the reduction in energy loss but very few referred to the need for particles to move in circular paths at high speed.

Question 04

4.1 This was generally well attempted, although a fair number of students did not gain marks due to incorrect extraction of data. This was also the case in 04.6. Some students wrote down the values but then failed to calculate two separate values of pV, simply writing $p_1V_1 = p_2V_2$ and making a conclusion that these were equal.

4.2 Most students recognized that the area was required to determine the work done. However, many failed to notice the false origin. Approximating the area to that of a triangle limited students to 1 mark.

The first law of thermodynamics in equation form on its own was insufficient. To score marks, students needed to define the terms.

The most common response was the increase in internal energy leading to an increase in temperature which enabled the students to access 2 marks. Students should be made aware that suggesting there is a change, rather than an increase or decrease, is not enough. They need to be more specific in their responses.

A significant number gave smooth, frictionless or longer as a required property of the cylinder.

4.7 A surprising number of students suggested plotting $\mathbf{ln}\,\boldsymbol{C}$ against $\mathbf{ln}\,\boldsymbol{V}$. There was a reasonable proportion of students who managed to correctly rearrange the relationship to that of an equation of a straight line, but then failed to suggest what should be plotted on each axis. Students should be reminded that when they are asked to explain how a term can be determined from the graph, the term needs to be the subject. In this case $\boldsymbol{b} = -\,\text{gradient}$ or $\boldsymbol{b} = -\frac{\mathbf{1}}{\text{gradient}}$.

Question 05

5.1 $\mathbf{3.01} \times \mathbf{10^{-3}}$ was a common response as students failed to use $\boldsymbol{T} = \boldsymbol{Fr}$. The use of diameter instead of radius was not penalized in marking point (mp)1. A large number of students were able to access mp2.

5.4 Most students were able to determine α. However, many tried to work out the ratio of diameters A and C rather than A and B.

5.5 Many referred to the torque being zero after 2.3 s.

Question 06

6.1 This was generally well attempted.

In 06.2, a large number simply referred to the two values being equal to the maximum voltage and the maximum current without any further comment.

The most common error in 06.4 was for students to use $\frac{0.54}{3.2}$ rather than $\frac{0.65}{3.5}$, thereby limiting themselves to 1 mark.

6.5 Any valid comparison was allowed, including: comparing the ratio of intensities to the ratio of the square of the distances, comparing 2 values of Ir^2 or comparing the power at the two distances.

The alternative mp2 based on comparing the order of magnitude of distance change with the number of significant figures in the distance was rarely seen.

Section B

The students performed well on most of the questions. Question 19 was the exception, where only just under 30% of students gave the correct answer and where option A proved to be the biggest distractor.